Logbook for week beginning 26/10/2026

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I had a meeting with Dr. Seeburger. We discussed the degraded synthetic spectra produced last week and we considered whether the exact resolution of the synthetic spectra was needed in the degradation process due to the observed spectrum's much larger σ. We also discussed the steps for this week which included created a grid plot of $\log(g)$ vs T_{eff} for the synthetic spectra and to degrade more of them, with the hopes of potentially calculating some radial velocities.

I then created the grid spectra by using the file names of the PHOENIX synthetic spectra to find their $\log(g)$ and T_{eff} and then plotting that to produce the grid

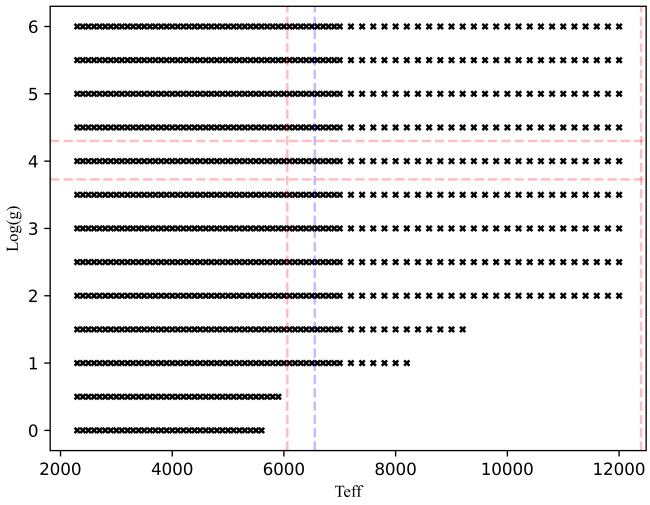


Figure 1: Grid of $\log(g)$ against T_{eff} for PHOENIX synthetic spectra. The red lines indicate the maximum and minimum values obtained from Gaia, while the blue line indicates the largest value when excluding source 4295254307232999296, which has $T_{\text{phot}} = 12395\,\text{K}$.

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I then wrote a script to take the $\log(g)$ and T_{eff} values for each candidate (every candidate has only phot, spec or none for both) and identify the synthetic spectra on the grid that is one $\log(g)$ and T_{eff} step above and below the candidates value. These were then downloaded from the PHOENIX library using `Phoenix4All` (https://github.com/taurex-space/phoenix4all). These were then degraded and converted from vacuum to air using the method of Ciddor, 1996. These spectra were then plotted on the non-degraded spectra. ‘

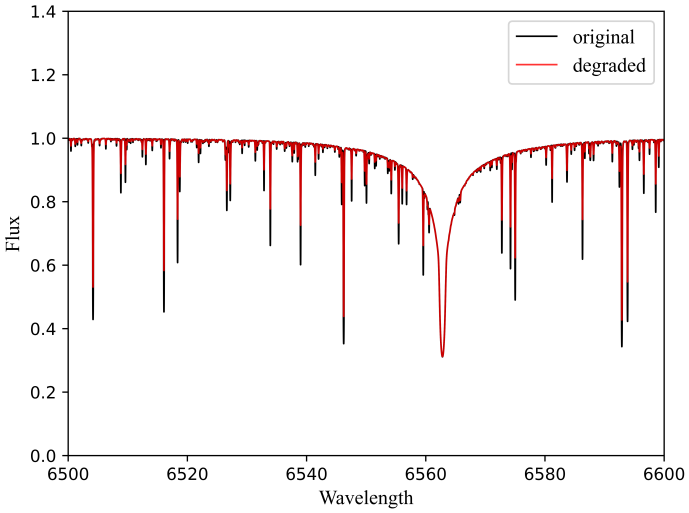


Figure 2: PHOENIX synthetic spectrum with $T_{\text{eff}} = 6000\,\text{K}$, $\log(g) = 4.00$ and [M/H] = 0.0, plotted between 650 and 660 nm. The spectrum assumes a resolving power of $R = 500{,}000$.

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Subtracted the continuum from the normalised spectra and implemented a 5% cosine bell, using the method from Tonry & David 1979, using the `tukey` function from `scipy`.

I then began to try and perform some cross correlation using the Fast Fourier Transform method. To do this, I used `crosscorrRV` from `PyAstronomy`, this successfully produced Correlation against Radial Velocity plots for all candidates for which synthetic spectra were downloaded and processed. For the candidates without T_{eff} and $\log(g)$ estimations from Gaia, they were ran against all synthetic spectra. 4295254307232999296, which exhibited double peaked Hα emission, did not look to be as expected against its synthetic spectra of $T_{eff} \approx 12000\,\text{K}$ and so I ran it against the $T_{eff} \approx 6000\,\text{K}$ spectra, which produced much more reasonable results. This indicates that the Gaia estimation for temperature for this target was inaccurate. Two of the candidates showed two strong peaks in the radial velocity plots, with 4239611700214875904 showing one strong and one weak peak.

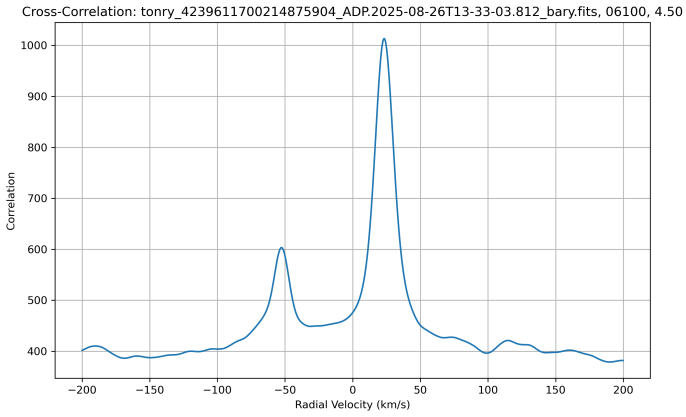


Figure 3: Cross correlation spectrum of observation ADP.2025-08-26T13-33-03.812 of 4239611700214875904 against PHOENIX synthetic spectra of $T_{eff} \approx 6100\,\text{K}$ and $\log(g) = 4.5$, [M/H] = 0.0.

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Began working on project introduction assignment. Current outline:

- What are (compact) binary star systems? What causes them to form and evolve? What are imposters?
- Why are they so hard to find? Why are they scientifically interesting? Why do we expect large populations to exist? Why do our observations not match expectations?
- How do we identify candidates? How do we analyse spectroscopic data? How do we distinguish genuine objects from imposters? How do we determine physical properties of them?
- So, what can they tell us about stellar evolution? So, what can they tell us about populations and formation? So, what could they tell us about other scientific implications?
- But, what have previous studies told us about them? But, what is currently happening in studies? But, what do we not currently know about them?
- Therefore, what does this study aim to achieve?

Weekly summary

Created the synthetic spectra grid as tasked in the meeting. Used this grid to identify up to four synthetic spectra for each candidate. These spectra were all degraded using the same method as the observed spectra, and corrections applied based on Tonry & Davis, 1979. These were then cross correlation using the `crosscorrRV` function from `PyAstronomy`, using an interpolation method, though I need to improve my understanding of this method or change it to a FFT version.

I also fixed bugs such as one caused by the addition of the Tonry & Davis method, where continuum subtracting the spectra was done twice on synthetic spectra causing the cross correlation to be inverted.

Corrections from previous week(s)

- Degraded spectra now include conversion for vacuum to air.

Files produced

- Updated `ADQL_test.py` to include cosine bell and subtract continuum.
- Updated `Synthetic_test.py` to create grid plot, download the 4 nearest synthetic spectra based on the grid plot and include cosine bell and subtract continuum.
- Created `cross_corr_test.py` to run cross correlation between observed and synthetic spectra.
- Produced cross correlation plots for all observations against all relevant synthetic spectra, including overlaid plots.

Tier 2 AI Policy:

I used Claude AI for bug fixing and planning code. I critically evaluated all outputs and remain responsible for the submitted work.